

Parametric Distributions

PSE 3.1

In some cases, a researcher may know (or be willing to assume that they know) the distribution of a random variable. More commonly, a researcher might be willing to assume that they know the distribution of a random variable up to a finite number of **parameters**, where the parameters allow for some flexibility in the shape of the distribution. In this section, we will briefly cover some of the most common/useful parametric distributions (though there are a number of other distributions mentioned in this chapter in the textbook that are worth mentioning).

Bernoulli Distribution

PSE 3.2

A random variable that follows a **Bernoulli distribution** takes either the value 0 or 1 with some probability p . An example is whether or not a particular person is employed. In particular, we can write

$$\begin{aligned}P(X = 1) &= p \\P(X = 0) &= 1 - p\end{aligned}$$

which fully summarizes the distribution of X . In practice, in some cases p might be known (e.g., for flipping a coin, $p = 0.5$), but in most interesting cases you would need to somehow estimate p (we'll return to this sort of issue later). It is also useful to have a single, full expression for the pmf of a Bernoulli random variable, which is given by

$$P(X = x) = p^x(1 - p)^{(1-x)}$$

In particular, just try plugging in $x = 1$ and $x = 0$ here and you will get $P(X = 1) = p$ and $P(X = 0) = (1 - p)$ as above.

Two interesting properties of Bernoulli random variables are:

1. $\mathbb{E}[X] = p$, in words: the expected value of X equals p
2. $\text{var}(X) = p(1 - p)$ in words: the variance of X equals $p(1 - p)$.

Here is a proof of the first property, and I'll leave proving the second property as an exercise.

$$\begin{aligned}\mathbb{E}[X] &= \sum_{x \in \{0,1\}} xP(X = x) \\&= 0(1 - p) + 1p \\&= p\end{aligned}$$

Binomial Distribution

A **Binomial** random variable counts the number of successes in n independent trials, each of which succeeds with the same probability p . Formally, if $X_1, \dots, X_n$ are iid Bernoulli(p) random variables (e.g., $X_i = 1$ if trial i is a “success” and 0 otherwise), then $X = X_1 + \dots + X_n$ follows a **Binomial** distribution, written $X \sim \text{Binomial}(n, p)$. For example, if you survey n people and record whether or not each one is employed, the total number of employed people in the sample is Binomial(n, p), where p is the (population) employment rate.

X is discrete with support $\{0, 1, \dots, n\}$, and its pmf is

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

where $\binom{n}{x}$ is the number of ways to choose which x of the n trials were successes (recall combinations from the Counting section of the first set of notes). A figure with the pmf of a Binomial random variable for a few different values of p is shown below.

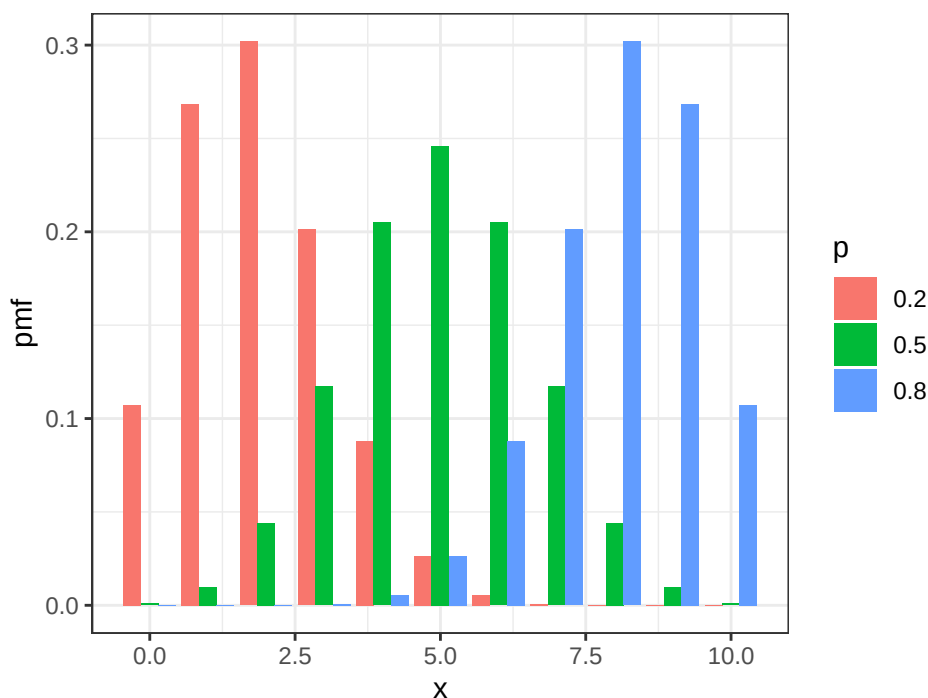


Figure 1: pmf of Binomial(10, p) for a few values of p

Because X is just a sum of n iid Bernoulli(p) random variables, we get $\mathbb{E}[X]$ for free from linearity of expectation (no new calculation needed):

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i] = \sum_{i=1}^n p = np$$

Similarly, since X_i are independent, it follows that

$$\text{var}(X) = \text{var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{var}(X_i) = \sum_{i=1}^n p(1-p) = np(1-p).$$

Poisson Distribution

A **Poisson** random variable counts the number of times some event occurs over a fixed interval of time or space, when those events occur independently at a constant average rate $\lambda > 0$. Examples of random variables that might follow a Poisson distribution include the number of patents a firm files in a given year, the number of customers arriving at a store in an hour, or the number of workplace accidents at a plant in a given month. We write $X \sim \text{Poisson}(\lambda)$.

X is discrete with support $\{0, 1, 2, \dots\}$ (unlike the Binomial, there is no upper bound), and its pmf is

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

The pmf of a Poisson random variable for a few different values of λ is shown below.

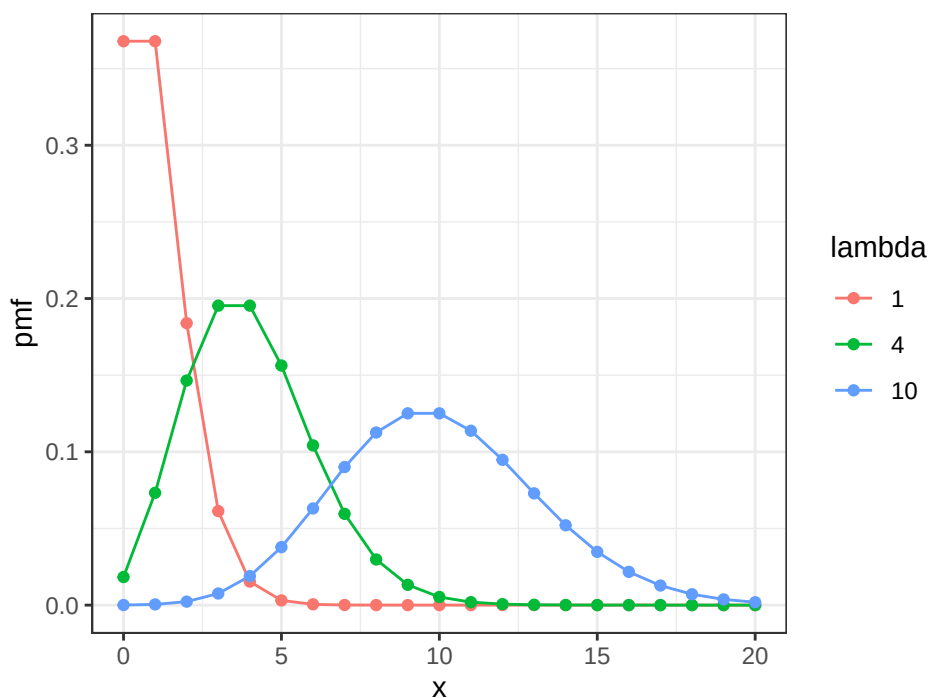


Figure 2: pmf of $\text{Poisson}(\text{lambda})$ for a few values of lambda

It turns out that $\mathbb{E}[X] = \lambda$ and $\text{var}(X) = \lambda$ (interestingly, the mean and variance are the same number for a Poisson random variable). Deriving $\mathbb{E}[X]$ isn't conceptually hard, but it takes a bit of algebra that I will omit here.

Uniform Distribution

A **Uniform** random variable is the natural continuous analogue of “all outcomes are equally likely” — X is equally likely to fall anywhere in some interval $[a, b]$. We write $X \sim \text{Uniform}[a, b]$. We already used this distribution in a couple of earlier examples.

X is continuous with support $[a, b]$, and its pdf is

$$f_X(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

(and $f_X(x) = 0$ outside of $[a, b]$).

```
unif_df <- data.frame(a = c(0, 1, 0), b = c(1, 3, 3))
unif_df$height <- 1 / (unif_df$b - unif_df$a)
unif_df$interval <- factor(paste0("U[", unif_df$a, ",", unif_df$b, "]"))

ggplot(data = unif_df) +
  geom_segment(aes(x = a, xend = b, y = height, yend = height, color = interval),
    linewidth = 1) +
  xlab("x") +
  ylab("pdf") +
  labs(color = NULL) +
  ylim(c(0,1)) +
  theme_bw()
```

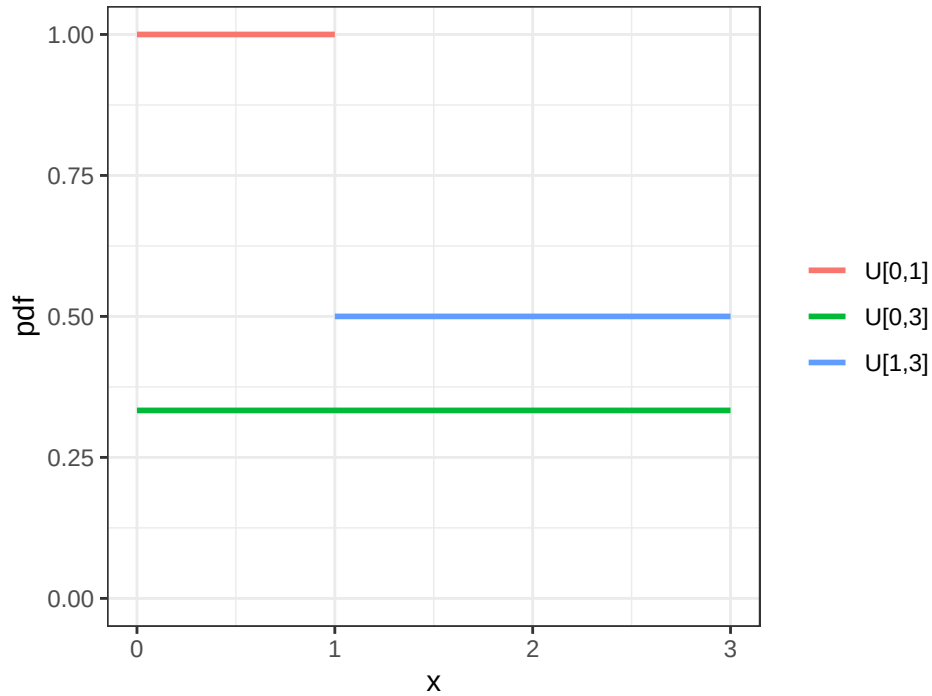


Figure 3: pdf of Uniform[a,b] for a few choices of a and b

By symmetry, $\mathbb{E}[X]$ is just the midpoint of the interval, $\mathbb{E}[X] = (a + b)/2$ (you can check this by direct integration). Similarly, $\text{var}(X) = (b - a)^2/12$.

Exponential Distribution

An **Exponential** random variable is a continuously distributed random variable that only takes positive values. An example of an economic variable that could follow an exponential distribution is unemployment duration. We write $X \sim \exp(\lambda)$.

X is continuous with support $[0, \infty)$, and its pdf is

$$f_X(x) = \frac{1}{\lambda} \exp(-x/\lambda), \quad x \geq 0, \lambda > 0$$

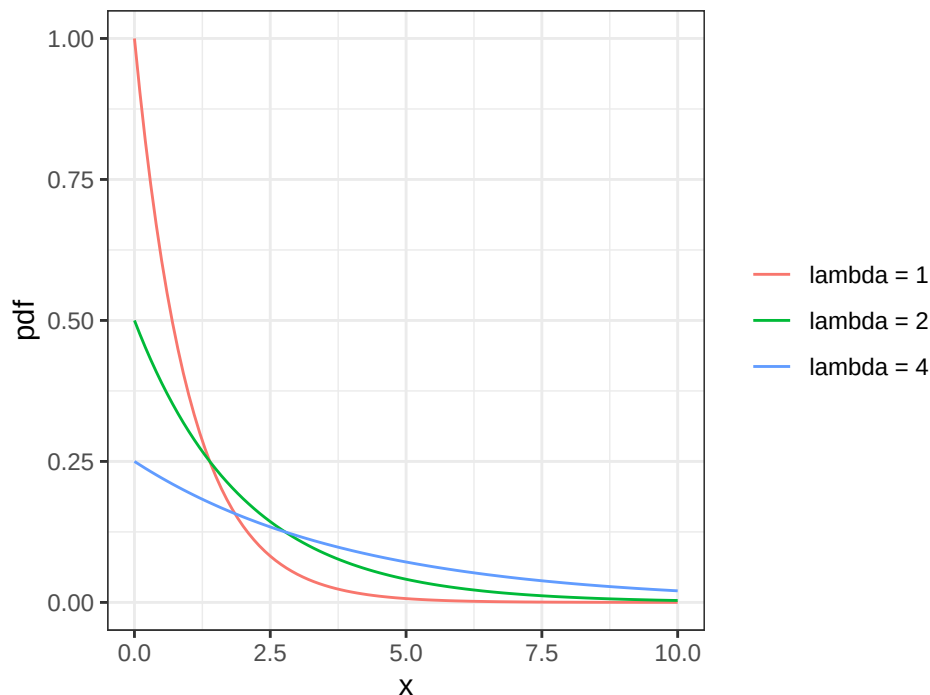


Figure 4: pdf of $\text{Exponential}(\text{lambda})$ for a few values of lambda

We already derived $\mathbb{E}[X] = \lambda$ for this distribution (using integration by parts) as a worked example in the previous set of notes. It also turns out that $\text{var}(X) = \lambda^2$.

Normal Distribution

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Arguably the most important parametric distribution is the normal distribution. We will soon see that random variables following a standard normal distribution show up naturally in statistics due to the central limit theorem.

A random variable X that follows a normal distribution is continuous and the normal distribution is indexed by two parameters μ (its mean) and σ^2 (its variance). It is common to write $X \sim \mathcal{N}(\mu, \sigma^2)$ for a random variable that follow a normal distribution. The pdf of X is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

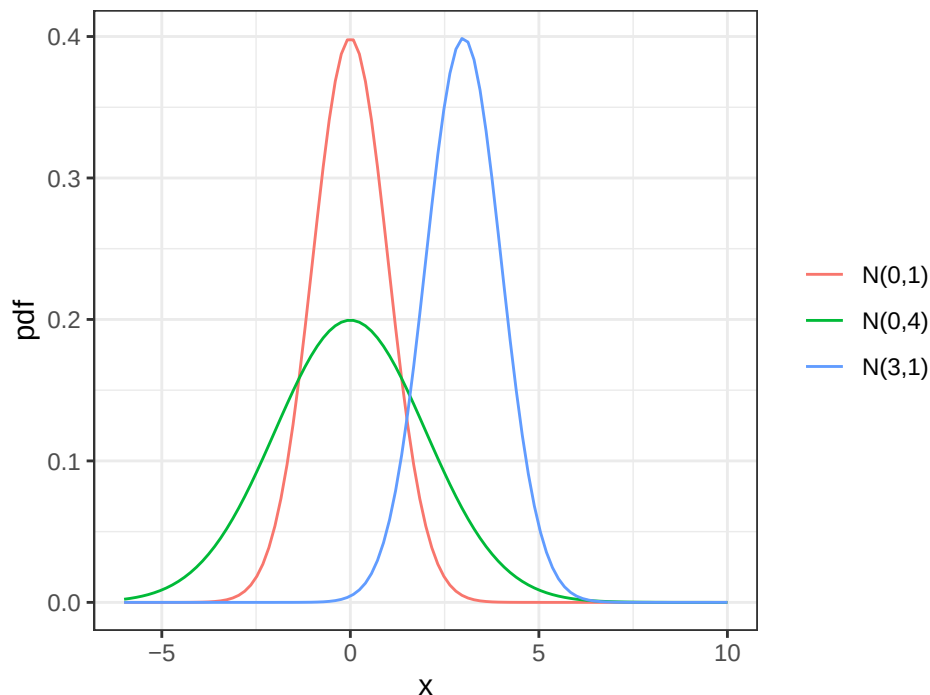


Figure 5: pdf of $\text{Normal}(\mu, \sigma^2)$ for a few parameter values

A special case (but important case) is when $\mu = 0$ and $\sigma^2 = 1$. In this case, X is said to follow a **standard normal distribution**.

Another useful property of normal distributions is stated in the following proposition

Proposition: Suppose that $X \sim \mathcal{N}(\mu, \sigma^2)$ and we define $Y = a + bX$ for some $a, b \in \mathbb{R}$. Then Y is also normally distributed as $N(a + b\mu, b^2\sigma^2)$.

A useful consequence of the previous proposition is that, when $X \sim \mathcal{N}(\mu, \sigma^2)$, it can be “standardized” by considering the transformed random variable $Z = (X - \mu)/\sigma$. This sort of transformation will be useful to us in the statistics portion of the course.

There is also some specialized notation that is worth mentioning for standard normal random variables. The cdf of a standard normal distribution is often denoted by Φ , and the pdf is often denoted by ϕ .

Next, if $X \sim \mathcal{N}(\mu, \sigma^2)$, then its moment generating function is given by $M_X(t) = \exp(\mu t + \sigma^2 t^2/2)$. For a standard normal random variable, $Z \sim \mathcal{N}(0, 1)$, its moment generating function is given by $M_Z(t) = \exp(t^2/2)$.

Normal distributions arise frequently in the context of statistical inference. From your undergraduate statistics class, you may be familiar with the idea of comparing some test statistic to a critical value that comes from a normal distribution. The next table comes from Table 5.1 in the textbook and provides the a summary of the cdf of a standard normal distribution (you’ll notice that some of the values of “x” in the table correspond to critical values that you may be familiar with).

Table 5.1: Normal Probabilities and Quantiles

	$\mathbb{P}[Z \leq x]$	$\mathbb{P}[Z > x]$	$\mathbb{P}[Z > x]$
$x = 0.00$	0.50	0.50	1.00
$x = 1.00$	0.84	0.16	0.32
$x = 1.65$	0.950	0.050	0.100
$x = 1.96$	0.975	0.025	0.050
$x = 2.00$	0.977	0.023	0.046
$x = 2.33$	0.990	0.010	0.020
$x = 2.58$	0.995	0.005	0.010

Chi-square Distribution

The **Chi-square** distribution arises directly from the normal distribution: if $Z_1, \dots, Z_k$ are iid standard normal random variables, then

$$X = Z_1^2 + Z_2^2 + \dots + Z_k^2$$

follows a **Chi-square distribution with k degrees of freedom**, written $X \sim \chi_k^2$. This distribution shows up throughout the statistics part of the course, particularly, for testing multiple hypotheses simultaneously.

X is continuous with support $[0, \infty)$. You do not necessarily need to know its pdf, but I'll mention it here for completeness (it is complicated and involves the Gamma function).

$$f_X(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}, \quad x > 0$$

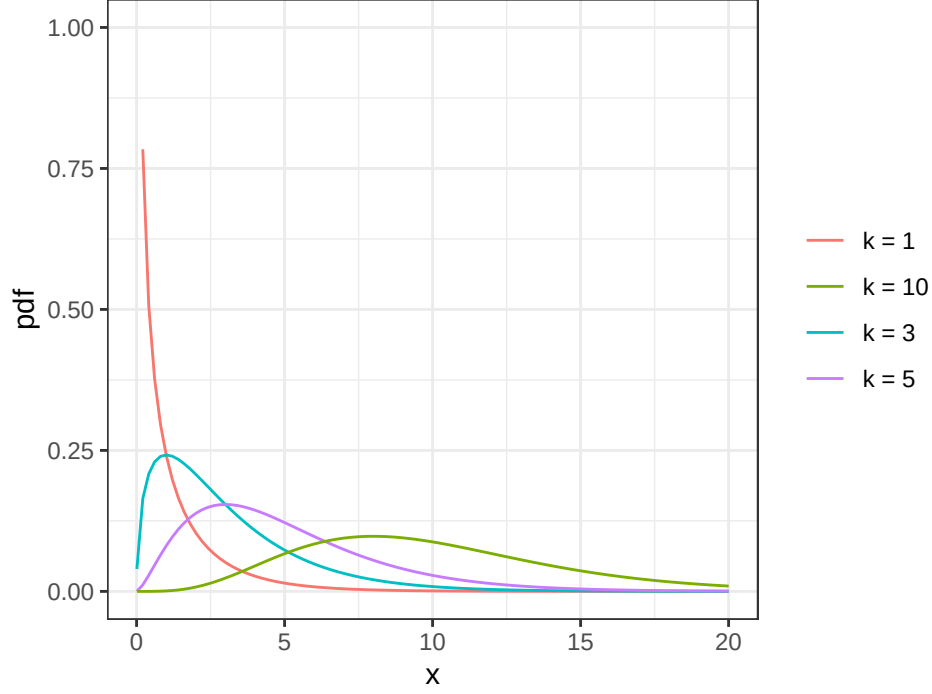


Figure 6: pdf of Chi-square(k) for a few degrees of freedom

Unlike the pdf, $\mathbb{E}[X]$ is easy to get directly from the construction above, reusing a fact we already know: since $Z_i \sim \mathcal{N}(0, 1)$, $\mathbb{E}[Z_i^2] = \text{var}(Z_i) = 1$. So, by linearity,

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^k Z_i^2\right] = \sum_{i=1}^k \mathbb{E}[Z_i^2] = k$$

It also turns out that $\text{var}(X) = 2k$, though the calculation is more involved, and we will omit it.

Side-Comment: A closely related distribution to a normal distribution is the **t-distribution**. A random variable X follows a t-distribution if it is the ratio of a standard normal random variable and the square root of an independent Chi-square random variable divided by its degrees of freedom, i.e., if $Z \sim \mathcal{N}(0, 1)$ and $U \sim \chi_k^2$ are independent, then $\frac{Z}{\sqrt{U/k}} \sim t_k$, i.e., a t -distribution with k degrees of freedom. This may seem like a strange (and oddly specific) construction, but it turns out that it arises naturally in the context of hypothesis testing, where a t -statistic follows a t -distribution when the (i) null hypothesis is true, (ii) the data are iid normal, and (iii) the variance of the population is unknown and must be estimated from the data. The t -distribution is similar to a standard normal distribution, but with heavier tails (the smaller the degrees of freedom, the heavier the tails). As the degrees of freedom increases, the t -distribution approaches a standard normal distribution (to see this, notice that $U/k \rightarrow 1$ as $k \rightarrow \infty$ because it is the average of k independent variables, each with mean 1).

Side-Comment: A closely related distribution to the chi-squared distributioin is the F -**distribution**. A random variable X follows an F -distribution if it is the ratio of two independent Chi-square random variables, each divided by its own degrees of freedom, i.e., if $U \sim \chi_{k_1}^2$ and $V \sim \chi_{k_2}^2$ are independent, then $\frac{U/k_1}{V/k_2} \sim F_{k_1, k_2}$. If you are used to using F-tests from another course, they are very closely related to chi-squared tests. The typical F-test involves looking at $F_{q, n-q}$ distribution, with q being the number of restrictions being tested. But note that, when X follows an $F_{q, n-q}$, qX follows a χ_q^2 distribution as long as n is large.

R Programming

R has built-in functions for every distribution covered in these notes. They follow a common naming convention: a one-letter prefix indicating *which* quantity you want, followed by a short name for the distribution. The four prefixes are

- **d** — the pdf (or the pmf, for a discrete distribution)
- **p** — the cdf, i.e., $P(X \leq x)$
- **q** — the quantile function, i.e., the inverse of the cdf; this is what you use to get critical values
- **r** — random draws from the distribution

For example, **dnorm** gives the normal pdf, **pnorm** the normal cdf, **qnorm** normal quantiles, and **rnorm** normal random draws. The table below collects the R name for each distribution along with the arguments that set its parameters — attach any of the four prefixes to the name in the third column.

Distribution	Our notation	R name	Setting the parameters
Bernoulli	$\text{Bernoulli}(p)$	binom	size = 1, prob = p
Binomial	$\text{Binomial}(n, p)$	binom	size = n, prob = p
Poisson	$\text{Poisson}(\lambda)$	pois	lambda
Uniform	$\text{Uniform}[a, b]$	unif	min = a, max = b
Exponential	$\text{Exp}(\lambda)$	exp	rate = 1/lambda
Normal	$\mathcal{N}(\mu, \sigma^2)$	norm	mean = mu, sd = sigma
Chi-square	χ_k^2	chisq	df = k
t	t_k	t	df = k
F	F_{k_1, k_2}	f	df1 = k1, df2 = k2

Notice that R has no separate Bernoulli function — a Bernoulli random variable is just a Binomial with **size** = 1.

The following code illustrates all four prefixes for a standard normal distribution.

```
dnorm(0)      # pdf evaluated at 0
```

```
[1] 0.3989423
```

```
pnorm(1.96)   # P(Z <= 1.96)
```

```
[1] 0.9750021
```

```
qnorm(0.975)  # the 97.5th percentile -- a familiar critical value
```

```
[1] 1.959964
```

```
set.seed(1234) # makes the random draws reproducible  
rnorm(3)       # three random draws
```

```
[1] -1.2070657  0.2774292  1.0844412
```

Side-Comment: Two of the distributions above are parameterized differently in R than in these notes, and mixing them up is a common source of hard-to-spot errors (the code runs fine, it just answers a different question than the one you meant to ask).

First, we wrote the exponential pdf as $f_X(x) = (1/\lambda) \exp(-x/\lambda)$, so that λ is the *mean* of the distribution. R instead parameterizes it by the *rate*, which is $1/\lambda$. So a mean- λ exponential is `dexp(x, rate = 1/lambda)`, not `dexp(x, lambda)`.

Second, we write a normal distribution as $\mathcal{N}(\mu, \sigma^2)$, in terms of its *variance*. R's `sd` argument is the *standard deviation*. So $\mathcal{N}(0, 4)$ is `dnorm(x, mean = 0, sd = 2)`, not `sd = 4`.